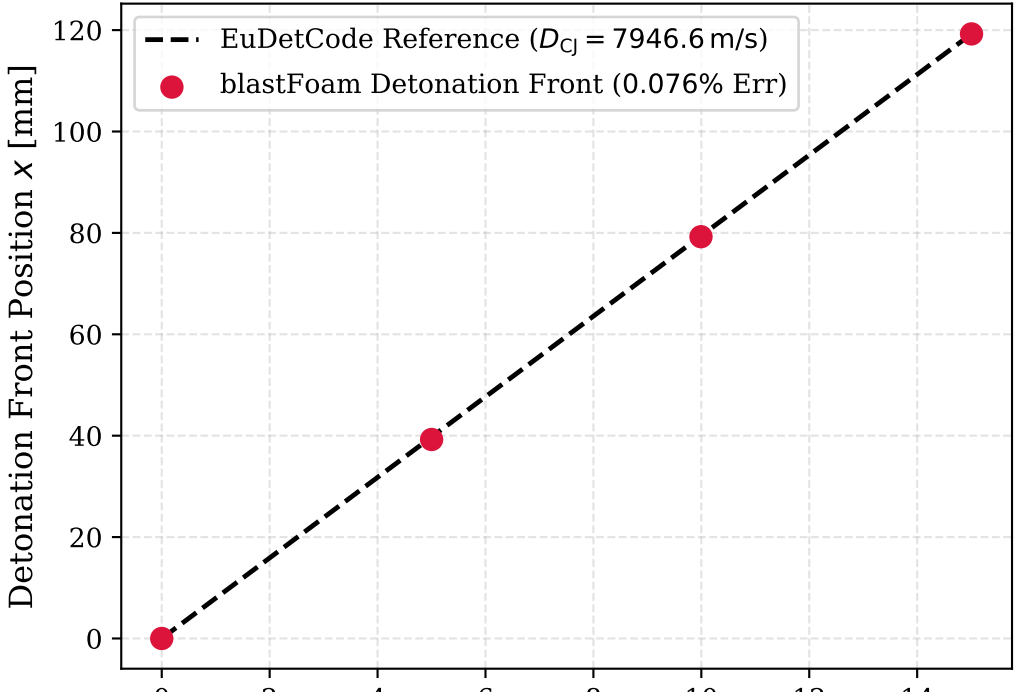
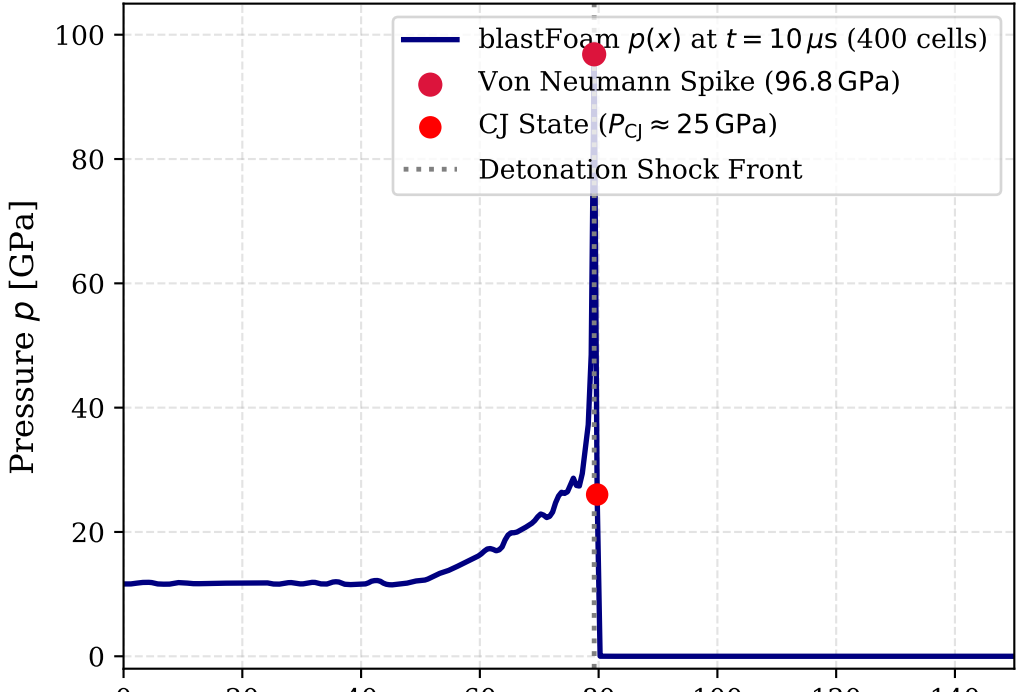

(a) 1D Planar Detonation Propagation
EuDetCode Reference ($D_{CJ} = 7946.6$ m/s)
blastFoam Detonation Front (0.076% Err)
Detonation Front Position x [mm]
Time t [μs]
0
20
40
60
80
100
120
0
2
4
6
8
10
12
14


(b) blastFoam Taylor Wave & Detonation Spike
blastFoam $p(x)$ at $t = 10\,\mu$s (400 cells)
Von Neumann Spike (96.8 GPa)
CJ State ($P_{CJ} \approx 25$ GPa)
Detonation Shock Front
Pressure p [GPa]
Distance from Initiation Wall x [mm]
0
20
40
60
80
100
120
140